

Annotated
Version

MACHINE LEARNING COURSE · CS-433

Regularization

Ridge Regression and Lasso

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Motivation

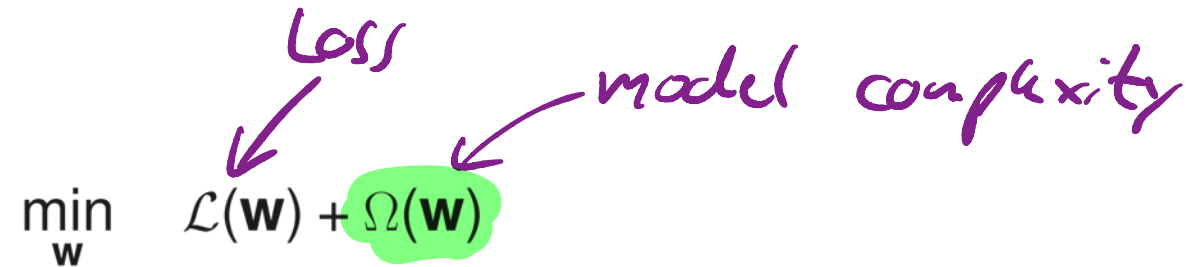
We have seen that by augmenting the feature vector we can make linear models as powerful as we want. Unfortunately this leads to the problem of overfitting.

Regularization is a way to mitigate this undesirable behavior.

We will discuss regularization in the context of linear models, but the same principle applies also to more complex models such as neural nets.

Regularization

Through [regularization](#), we can penalize complex models and favor simpler ones:

$$\min_{\mathbf{w}} \mathcal{L}(\mathbf{w}) + \Omega(\mathbf{w})$$


The second term Ω is a [regularizer](#), measuring the complexity of the model given by w .

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L_2 -regularization: ridge regression

Penalize the Euclidean norm of the weights.

L_2 -regularization

The most frequently used regularizer is the standard Euclidean norm (L_2 -norm), that is

$$\Omega(\mathbf{w}) = \lambda \|\mathbf{w}\|_2^2 \quad \text{where} \quad \|\mathbf{w}\|_2^2 = \sum_i w_i^2$$

Here the main effect is that large model weights w_i will be penalized (avoided), since we consider them “unlikely”, while small ones are ok.

When L is MSE, this is called **ridge regression**:

$$\min_{\mathbf{w}} \underbrace{\frac{1}{2N} \sum_{n=1}^N [y_n - \mathbf{x}_n^\top \mathbf{w}]^2}_{\text{loss}(\mathbf{w})} + \underbrace{\lambda \|\mathbf{w}\|_2^2}_{\Omega(\mathbf{w})}$$

$$-\nabla \Omega = -2\lambda \mathbf{w}$$

weight decay

Least squares is a special case of this: set $\lambda := 0$.

Explicit solution for w

Differentiating and setting to zero:

$$\begin{aligned} \nabla \mathcal{L} &= -\frac{1}{N} X^T (y - Xw) \\ + \nabla \Omega &= 2\lambda w \\ \hline \nabla \rho & \stackrel{!}{=} 0 \end{aligned}$$

$$w_{\text{ridge}}^* = (X^T X + \lambda' I)^{-1} X^T y$$

(here for simpler notation $\lambda' / 2N = \lambda$)

$$\Leftrightarrow (X^T X + \lambda' I) w = X^T y$$

solve linear system

$$\rightarrow w^*$$

Ridge regression to fight ill-conditioning

before: $EV \geq 0$
after: $EV \geq \lambda' > 0$

The eigenvalues of $(\underline{X^T X} + \lambda' I)$ are all at least λ' , and so the inverse always exists.

This is also referred to as *lifting the eigenvalues*.

Proof: Write the eigenvalue decomposition of $X^T X$ as USU^T . We then have

$$\begin{aligned} X^T X + \lambda' I &= USU^T + \lambda' UIU^T \\ &= \underline{U[S + \lambda' I]U^T} \end{aligned}$$

We see now that every eigenvalue is “lifted” by an amount λ' .



An alternative proof

optional

Recall that for a symmetric matrix A we can also compute eigenvalues by looking at the so-called Rayleigh ratio,

$$R(\mathbf{A}, \mathbf{v}) = \frac{\mathbf{v}^\top \mathbf{A} \mathbf{v}}{\mathbf{v}^\top \mathbf{v}}$$

If $\mathbf{v}$ is an eigenvector with eigenvalue λ then the Rayleigh coefficient indeed gives us λ . We can find the smallest and largest eigenvalue by minimizing and maximizing this coefficient. Applying this to the symmetric matrix $\mathbf{X}^\top \mathbf{X} + \lambda' \mathbf{I}$, for any vector $\mathbf{v}$ we have

$$\frac{\mathbf{v}^\top (\mathbf{X}^\top \mathbf{X} + \lambda' \mathbf{I}) \mathbf{v}}{\mathbf{v}^\top \mathbf{v}} \geq \frac{\lambda' \mathbf{v}^\top \mathbf{v}}{\mathbf{v}^\top \mathbf{v}} = \lambda'$$

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L_1 -regularization: the Lasso

A different norm, and a very different kind of solution.

L_1 -regularization: the Lasso

$$\|w\|_1 = \sum_{i=1}^D |w_i|$$

As an alternative measure of the complexity of the model, we can use a different norm. A very important case is the L_1 -norm, leading to L_1 -

regularization. In combination with the MSE cost function, this is known as the **Lasso**: *linear model*

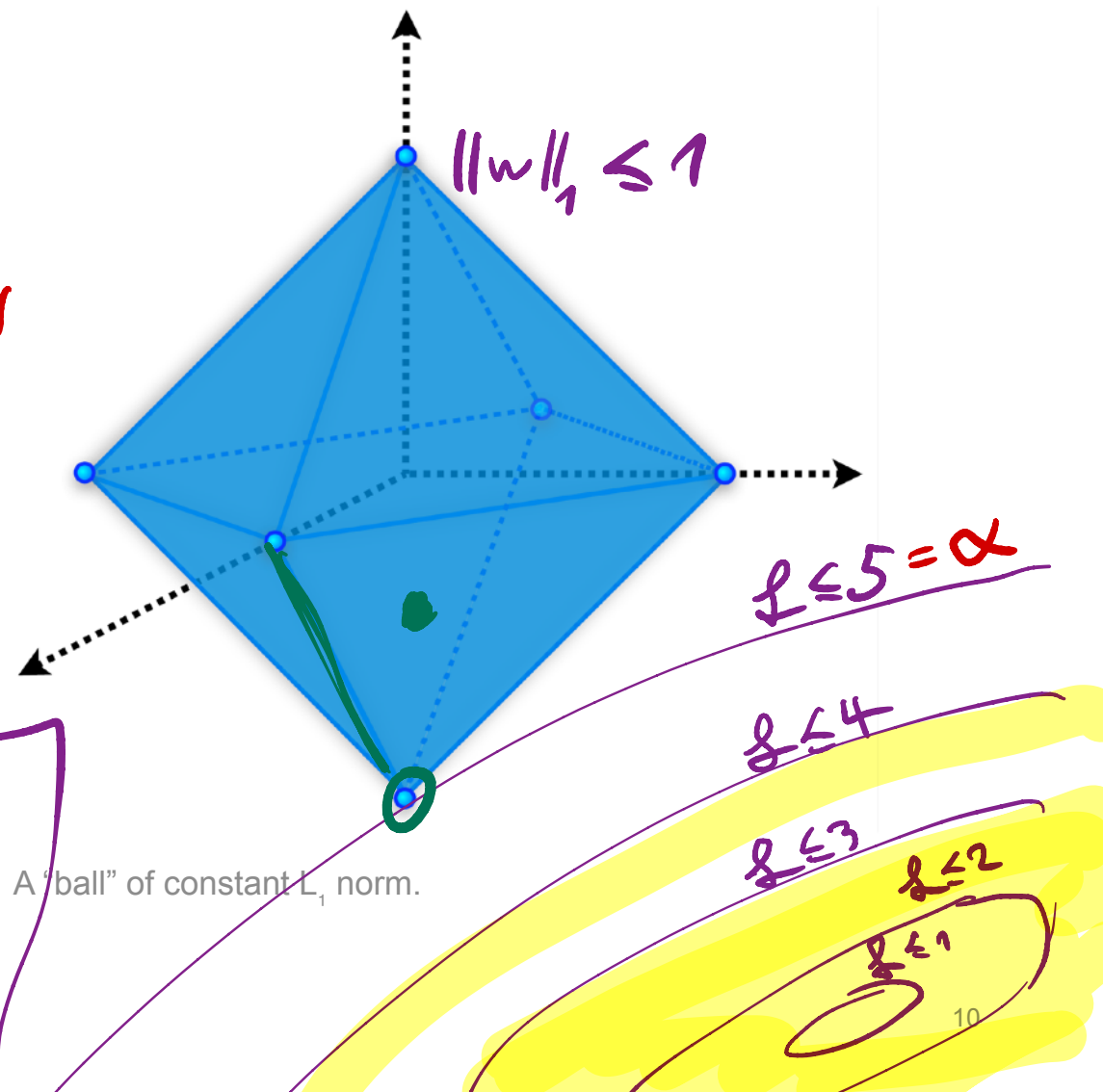
$$\min_w \frac{1}{2N} \sum_{n=1}^N [y_n - \mathbf{x}_n^\top \mathbf{w}]^2 + \lambda \|\mathbf{w}\|_1$$

$\mathcal{L}(w)$

where

$$\|\mathbf{w}\|_1 := \sum_i |w_i|$$

$$\begin{aligned} \min \mathcal{L}(w) \\ \text{s.t. } \|w\|_1 \leq c \end{aligned}$$



Why does L_1 give sparse solutions?

To keep things simple assume that $X^\top X$ is invertible. We claim that the set

$$\{ \mathbf{w} : \overbrace{\|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2}^{L(\mathbf{w})} = \alpha \}$$

← ellipse, level set

is an ellipsoid, and this ellipsoid simply scales around its origin as we change α .

We claim that for L_1 -regularization the optimum solution is likely going to be sparse (only has few non-zero components) compared to the case where we use L_2 -regularization.

The geometric argument

Assume that a genie tells you the L_1 -norm of the optimum solution. Draw the L_1 -ball with that norm value (think of 2D to visualize it). So now you know that the optimal point is somewhere on the surface of this “ball”.

Further, you know that there are ellipsoids, all with the same mean and rotation, that describe the equal-error surfaces incurred by the first term. The optimum solution is where the “smallest” of these ellipsoids just touches the L_1 -ball.

Due to the geometry of this ball, this point is more likely to be on one of the “corner” points. In turn, sparsity is desirable, since it leads to a “simple” model.

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Additional notes

Other regularizers, the constrained view, and the Bayesian reading.

Other types of regularization

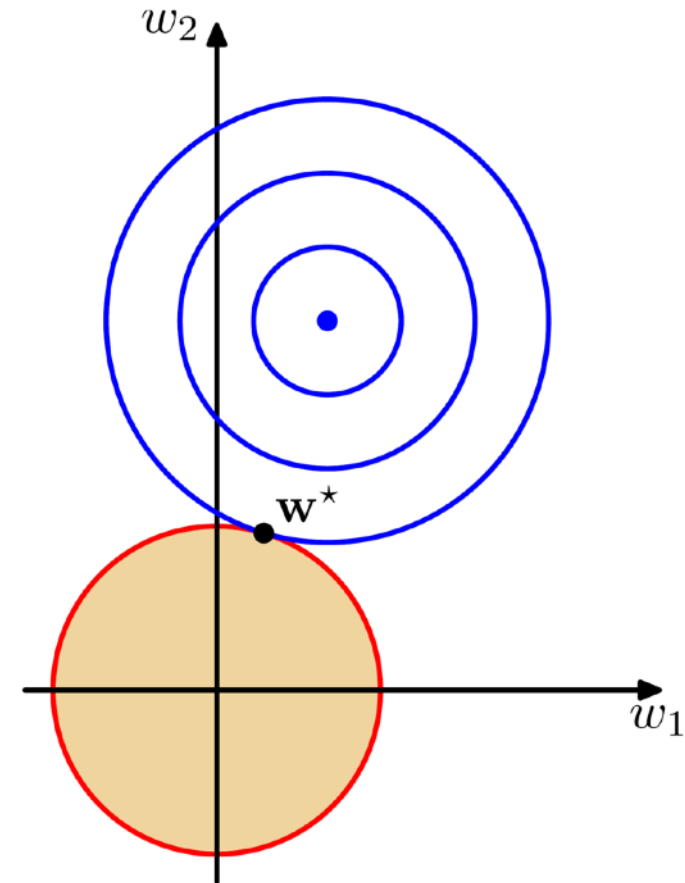
Popular methods such as [shrinkage](#), [dropout](#) and [weight decay](#) (in the context of neural networks), and [early stopping of the optimization](#) are all different forms of regularization.

Regularization as a constraint

Another view of regularization: the ridge regression formulation we have seen above is similar to the following constrained problem (for some $\tau > 0$).

$$\min_{\mathbf{w}} \quad \frac{1}{2N} \sum_{n=1}^N (y_n - \mathbf{x}_n^\top \mathbf{w})^2, \quad \text{such that } \|\mathbf{w}\|_2^2 \leq \tau$$

The picture illustrates this: the blue lines are the level sets of the MSE cost function, and the shaded disc is the constraint set.



Geometric interpretation of ridge regression.

The same view for the Lasso

For the case of using L_1 regularization (known as the [Lasso](#), when used with MSE) we analogously consider

$$\min_{\mathbf{w}} \quad \frac{1}{2N} \sum_{n=1}^N (y_n - \mathbf{x}_n^\top \mathbf{w})^2, \quad \text{such that } \|\mathbf{w}\|_1 \leq \tau$$

This forces some of the elements of w to be strictly 0 and therefore enforces sparsity in the model (some features will not be used since their coefficients are zero).

- Why does L_1 regularizer enforce sparsity? *Hint:* draw a picture similar to the above, and locate the optimal solution.
- Why is it good to have sparsity in the model? Is it going to be better than least-squares? When and why?

Least squares as maximum likelihood

Recall that classic *least-squares linear regression* can be interpreted as the [maximum likelihood estimator](#):

$$\begin{aligned}
 \mathbf{w}_{\text{lse}} &\stackrel{(a)}{=} \arg \min_{\mathbf{w}} -\log p(\mathbf{y}, \mathbf{X} \mid \mathbf{w}) \\
 &\stackrel{(b)}{=} \arg \min_{\mathbf{w}} -\log p(\mathbf{X} \mid \mathbf{w}) p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}) \\
 &\stackrel{(c)}{=} \arg \min_{\mathbf{w}} -\log p(\mathbf{X}) p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}) \\
 &\stackrel{(d)}{=} \arg \min_{\mathbf{w}} -\log p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}) \\
 &\stackrel{(e)}{=} \arg \min_{\mathbf{w}} -\log \left[\prod_{n=1}^N p(y_n \mid \mathbf{x}_n, \mathbf{w}) \right] \\
 &\stackrel{(f)}{=} \arg \min_{\mathbf{w}} -\log \left[\prod_{n=1}^N \mathcal{N}(y_n \mid \mathbf{x}_n^\top \mathbf{w}, \sigma^2) \right] \\
 &= \arg \min_{\mathbf{w}} \frac{1}{2\sigma^2} \sum_{n=1}^N (y_n - \mathbf{x}_n^\top \mathbf{w})^2
 \end{aligned}$$

(a) is the negative log-likelihood; (b) factors it; (c) uses that the inputs do not depend on $\mathbf{w}$; (d) drops the constant $p(\mathbf{X})$; (e) uses that the samples are iid; (f) uses the Gaussian noise assumption. The rest is calculus.

Ridge regression as MAP estimator

Ridge regression has a very similar interpretation. Now we start with the posterior $p(\mathbf{w} \mid \mathbf{X}, \mathbf{y})$ and choose the $\mathbf{w}$ that maximizes it — the [maximum-a-posteriori \(MAP\) estimate](#). We assume the components of the weight vector are iid Gaussians with mean zero and variance $1/\lambda$.

$$\begin{aligned}
 \mathbf{w}_{\text{ridge}} &= \arg \min_{\mathbf{w}} -\log p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) \\
 &\stackrel{(a)}{=} \arg \min_{\mathbf{w}} -\log \frac{p(\mathbf{y}, \mathbf{X} \mid \mathbf{w}) p(\mathbf{w})}{p(\mathbf{y}, \mathbf{X})} \\
 &\stackrel{(b,c)}{=} \arg \min_{\mathbf{w}} -\log p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}) p(\mathbf{w}) \\
 &= \arg \min_{\mathbf{w}} -\log \left[\mathcal{N}(\mathbf{w} \mid 0, \frac{1}{\lambda} \mathbf{I}) \prod_{n=1}^N \mathcal{N}(y_n \mid \mathbf{x}_n^\top \mathbf{w}, \sigma^2) \right] \\
 &= \arg \min_{\mathbf{w}} \sum_{n=1}^N \frac{1}{2\sigma^2} (y_n - \mathbf{x}_n^\top \mathbf{w})^2 + \frac{\lambda}{2} \|\mathbf{w}\|^2
 \end{aligned}$$

In step (a) we used Bayes' law. In steps (b) and (c) we eliminated quantities that do not depend on $\mathbf{w}$.

Regularization as prior information

Based on the previous derivation of ridge regression, we can more generally see regularization as encoding any kind of prior information we have.

It can thus be understood as a compressed form of data — in this sense, using regularization is equivalent to adding data, which helps reduce overfitting.